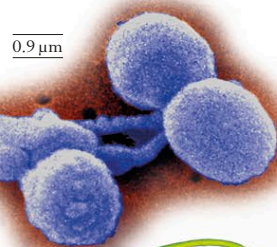


CLOSTRIDIUM BOTULINUM

This bacterium thrives where oxygen is limited or absent. Living in soil, it produces the neurotoxins that cause botulism, but which also have medical and cosmetic uses.

CLOSTRIDIUM TETANI

A soil organism that can grow in dead tissue in wounds or burns, *C. tetani* produces tetanospasmin, the neurotoxin that causes tetanus.



STREPTOCOCCUS PNEUMONIAE

This bacterium, present inside humans, can cause pneumonia. In children and the elderly, it is the leading cause of invasive infection that moves into all parts of the body.

2–3 μm

SALMONELLA ENTERICA

Belonging to the same family as *E. coli*, some subspecies of *Salmonella* bacteria cause gastroenteritis; others cause typhoid fever.



LACTOBACILLUS ACIDOPHILUS

Found in the intestine and vagina, this bacterium has nutritional and antimicrobial properties. It is used in probiotic drinks and supplements.

flagellum helps propel organism

1–3 μm

VIBRIO CHOLERAЕ

Highly mobile curved rods with a single flagellum at one end, this species secretes a potent enterotoxin that causes cholera.

NITROSOSPIRA SP.

Filling a vital ecological niche, this nitrifying soil bacteria oxidizes ammonia to form nitrites as part of the nitrogen cycle.

cell contents, or cytoplasm

1–3 μm



YERSINIA PESTIS

This species is the agent that causes bubonic plague. Transmitted to humans via fleas that live on rats, it causes up to 3,000 cases worldwide each year.

SHIGELLA DYSENTERIAE

This gut bacteria produces shiga toxin that can cause epidemic dysentery. As few as ten bacteria are needed to cause an infection.



STAPHYLOCOCCUS EPIDERMIDIS

This spherical bacterium can be found on the skin as part of normal flora, but can cause infections in immune-compromised patients.

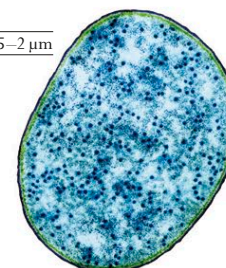
0.4–0.5 μm



FUSOBACTERIUM NUCLEATUM

This species makes its home in the human mouth, and forms a major component of plaque. It can also cause premature birth.

1.5–2 μm



PSYCHROBACTER URATIVORANS

This species is a psychrophile, or cryophile, which means it thrives at very low temperatures, thanks to natural antifreeze molecules in its cytoplasm within the cell membrane.

cell wall



NOSTOC SP.

This cyanobacteria forms gelatinous filaments in colonies that are hardy enough to survive from the poles to the tropics.

cell dividing

3–7 μm

DNA

pigmented membranes for photosynthesis

numerous rods of bacteria grouped together

ENTEROCOCCUS FAECALIS

Normally a harmless inhabitant of the human digestive tract and vagina, this species can invade wounds and is resistant to many antibiotics.

0.5–1 μm

